

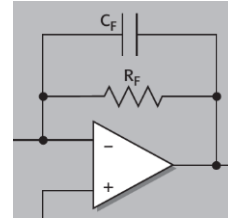
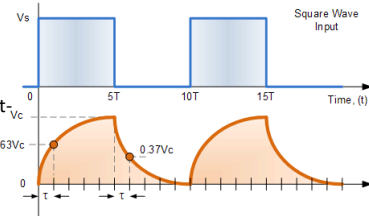
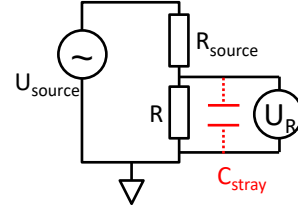
RC time constant

Cables and circuit elements always have a finite stray capacitance, which, together with the appropriate resistors in the circuit, forms an RC filter. Example: AC voltage source (U_{source}) is used to test resistance R in current generator mode. The circuit current is set with resistor $R_{\text{source}} \gg R$ (i.e. $I_{\text{source}} \approx U_{\text{source}}/R_{\text{source}}$), the voltage is measured at R . The resistor R and the long coax cable form a parallel RC circuit. Without the stray capacitance $U_R = I_{\text{source}} R$ voltage should drop on the resistor. However, if the absolute value of the stray capacitance's complex impedance becomes smaller than the resistance, $|Z_C| = 1/(\omega C_{\text{stray}}) < R$, most of the current flows through the capacitance and the measured U_R voltage strongly decreases. This means, that at frequency $2\pi f = \omega < 1/RC_{\text{stray}}$ the desired voltage is measured, while above the cut-off frequency $\omega = 1/RC_{\text{stray}}$ the measured voltage is cut-off, i.e. it becomes smaller as the frequency is increased.

The typical capacitance of a coax cable is 50-100pF/m, so with a 6m cable and $R = 3.3\text{k}\Omega$ (see experiment) we expect a cut-off at $f \sim 160\text{-}320\text{kHz}$.

If we drive the circuit with a square wave, we see a charge/discharge at the leading and trailing edges with a time constant of $\tau = RC_{\text{stray}}$

Another example: when measuring low currents, we usually use a current amplifier with a high feedback resistor. With $R_F = 1\text{G}\Omega$ feedback resistor and $C_F = 1\text{pF}$ stray capacitance, the instrument bandwidth is $f_{\text{max}} = 1/2\pi RC_F \sim 160\text{Hz}$, you cannot measure a signal with a higher frequency than this, i.e. you cannot measure small currents quickly. A significantly lower stray capacitance than this is not very feasible, but the stray capacitance can be somewhat compensated by clever tricks.



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Signal propagation in coaxial lines, telegrapher's equations

If the length of the cable is comparable to the wavelength of the electromagnetic wave propagating at a given frequency, then the description for low frequency circuits is no longer applicable!

Coaxial cable model: the outer (shield) wire is grounded. A dx long elementary segment of the inner conductor has a self-inductance coefficient $\tilde{L}dx$ and a resistance $\tilde{R}dx$. Furthermore, a dx long elementary segment of the inner conductor has a $\tilde{C}dx$ capacitance and $\tilde{G}dx$ conductance towards the shield. From these parameters the resistance of the inner wire and the conductance towards the shield is neglected ($\tilde{R} = 0, \tilde{G} = 0$). Due to the self-inductance, the voltage drop on the elementary section of wire dx is:

$$dU = -\tilde{L}dx \cdot \frac{dI}{dt},$$

while the variation of the current along a dx segment is determined by the charging and discharging of the capacitance towards the shield:

$$dI = -\tilde{C}dx \cdot \frac{dU}{dt}.$$

These can be used to write down the coupled differential equations on the current and voltage, the so-called telegrapher's equations:

$$\frac{\partial U(x,t)}{\partial x} = -\tilde{L} \cdot \frac{\partial I(x,t)}{\partial t}, \quad \frac{\partial I(x,t)}{\partial x} = -\tilde{C} \cdot \frac{\partial U(x,t)}{\partial t},$$

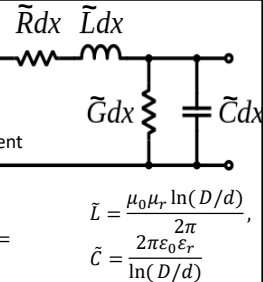
which simply give the following wave equations:

$$\frac{\partial^2 U(x,t)}{\partial t^2} = v_{\text{signal}}^2 \cdot \frac{\partial^2 U(x,t)}{\partial x^2}, \quad \frac{\partial^2 I(x,t)}{\partial t^2} = v_{\text{signal}}^2 \cdot \frac{\partial^2 I(x,t)}{\partial x^2},$$

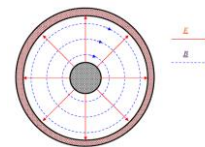
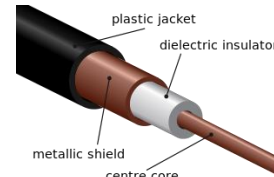
where v_{signal} is the propagation speed of the EM wave in the coaxial cable:

$$v_{\text{signal}} = \frac{1}{\sqrt{\tilde{L}\tilde{C}}}$$

the signal travels at approximately $\sim 2/3$ the speed of light in coaxial cables



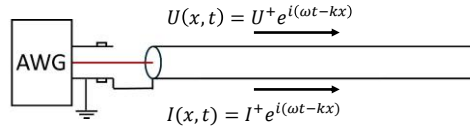
D and d are the diameters of the outer and inner wires



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Solution of the telegrapher's equation for waves with ω circular frequency

First, let's assume that a wave with ω circular frequency travels in an infinite transmission line in positive direction. In this case no reflection (i.e. no wave travelling in the negative direction) is considered:



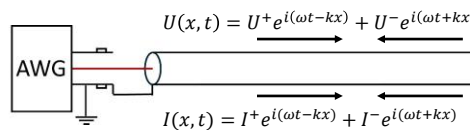
Substituting this to the first telegrapher's equation the relation between the voltage-wave and the current-wave is obtained:

$$-ikU^+ e^{i(\omega t - kx)} = -i\omega \tilde{L} I^+ e^{i(\omega t - kx)} \Rightarrow \frac{U^+}{I^+} = \tilde{L} \frac{\omega}{k} = \sqrt{\frac{\tilde{L}}{\tilde{C}}} = Z_0.$$

$v_{\text{signal}} = 1/\sqrt{\tilde{L}\tilde{C}}$

Z_0 is the so-called wave impedance. This quantity has the dimension of resistance describing the ratio of the voltage and current wave-amplitudes.

In the general case, both positive and negative direction waves propagate in the line:



Substituting these wavefunctions to the both

telegrapher's equation at the $x=0$ position the following relations are obtained:

$$\begin{aligned} -kU^+ + kU^- &= -\tilde{L}\omega(I^+ + I^-) \\ -kI^+ + kI^- &= -\tilde{C}\omega(U^+ + U^-) \end{aligned}$$

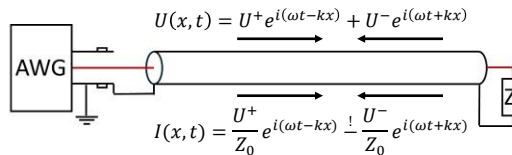
From these the ratio of the voltage and current amplitudes is obtained for both the positive and negative direction waves, mind the negative sign in the latter case:

$$I^+ = U^+ / Z_0, \quad I^- = -U^- / Z_0$$

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Reflections on a terminating resistance, impedance matching

Now, we connect the inner wire and the shield at the end of the coaxial line through a Z_L load impedance, which introduces reflections at the wire end ($x=0$ position). The arbitrary solutions for harmonic waves are shown on the figure, note that the previous results for the ratios of the current and voltage amplitudes are already applied.



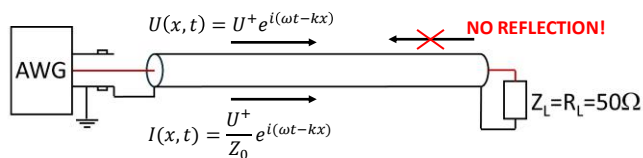
As a boundary condition, at the wire end the following equation should apply:

$$Z_L = \frac{U(x=0, t)}{I(x=0, t)} = \frac{U^+ + U^-}{\frac{U^+}{Z_0} - \frac{U^-}{Z_0}}$$

Rearranging this equation the ratio of the reflected voltage wave amplitude (U^-) and the incoming voltage wave amplitude (U^+) is obtained. This Γ coefficient describes the "reflectivity" of the Z_L terminating impedance:

$$\Gamma = \frac{U^-}{U^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Let us first assume, that the terminating impedance is the same as the wave-impedance, $Z_L = Z_0$. The most common coaxial wires in the lab have 50Ω wave impedance, i.e. we terminate the wire with a 50Ω resistor. In this case the reflection coefficient is zero, i.e. only the positive direction harmonic wave coming from the arbitrary waveform generator (AWG) travels in the wire. This is the case of impedance matching ($Z_L = Z_0$), which prevents reflections at the terminating resistor:



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Reflections on a terminating resistance: free end

$$\Gamma = \frac{U^-}{U^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Next, we assume, that the coaxial wire has a free end, i.e. the inner wire and the shield are disconnected ($Z_L = \infty$). In this case $\Gamma = 1$ reflection coefficient is obtained, i.e the entire wave is reflected ($U^- = U^+$):

$$\left(n + \frac{1}{2}\right) \cdot \frac{\lambda}{2} = \left(n + \frac{1}{2}\right) \cdot \frac{\pi}{k} = \left(n + \frac{1}{2}\right) \cdot \frac{\pi v_{\text{signal}}}{\omega} = \left(n + \frac{1}{2}\right) \cdot v_{\text{signal}} / (2f)$$

$$U(x, t) = U^+ e^{i(\omega t - kx)} + U^+ e^{i(\omega t + kx)} = 2U^+ e^{i\omega t} \cos(kx)$$

$$I(x, t) = \frac{U^+}{Z_0} e^{i(\omega t - kx)} - \frac{U^+}{Z_0} e^{i(\omega t + kx)} = -\frac{2U^+}{Z_0} e^{i\omega t} \sin(kx)$$

Note that here, the incoming and reflected voltage waves form a standing wave with a maximum voltage amplitude at the end of the wire, and cancellations at distances corresponding to half-integer multiples of the half wavelength. In contrast, the current-wave has cancellation at the wire end due to the infinite terminating impedance.

Note, that at low frequency ($kx \ll \pi/2$) the entire inner wire would oscillate according to the driving voltage ($2U^+ e^{i\omega t}$). Note, that a factor of 2 difference is obtained if the function generator with 50Ω output resistance drives an impedance-matched (50Ω wave impedance) coaxial wire, or if it drives an infinite impedance system.

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Reflections on a terminating resistance: short-circuit termination

$$\Gamma = \frac{U^-}{U^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Finally, we assume, that the coaxial wire-end is short-circuited ($Z_L = 0\Omega$). In this case $\Gamma = -1$ reflection coefficient is obtained, i.e the entire wave is reflected with a minus sign ($U^- = -U^+$):

$$n \cdot \frac{\lambda}{2} = n \cdot \frac{\pi}{k} = n \cdot \frac{\pi v_{\text{signal}}}{\omega} = n \cdot v_{\text{signal}} / (2f)$$

$$U(x, t) = U^+ e^{i(\omega t - kx)} - U^+ e^{i(\omega t + kx)} = -2U^+ e^{i\omega t} \sin(kx)$$

$$I(x, t) = \frac{U^+}{Z_0} e^{i(\omega t - kx)} + \frac{U^+}{Z_0} e^{i(\omega t + kx)} = \frac{2U^+}{Z_0} e^{i\omega t} \cos(kx)$$

Again, the incoming and reflected voltage waves form a standing wave, which has zero voltage at the wire end due to the short circuit, and cancellations are also observed at integer wavelength distances. Note, that here the current wave has maximum amplitude at the wire end.

Note, that at low frequency ($kx \ll \pi/2$) the entire inner wire would be at zero voltage due to the short circuit at the end.

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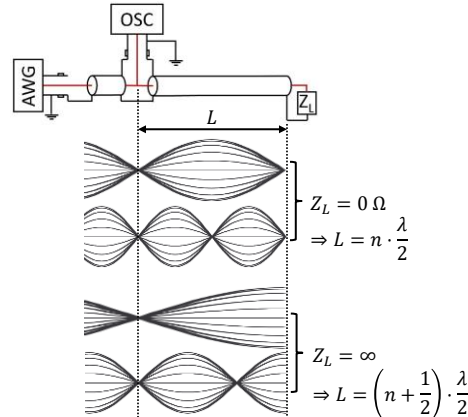
Experiments with transmission lines: measurement of propagation speed through standing wave tests

Experiment: testing cable-end reflection with harmonic excitation. The sinusoidal output of a signal generator is connected to the input of an oscilloscope by a short coaxial cable, and a long ($L \approx 6\text{m}$) coaxial cable is connected to the input of the oscilloscope by a T-divider, the end of which is first terminated with 50Ω . Changing the frequency (f) does not change the amplitude of the signal measured by the oscilloscope in this case.

If the far end of the long coax is terminated with zero resistance (short circuit), oscillations are seen as a function of frequency with dedicated cancellations of the signal. The lowest frequency cancellation corresponds to a standing wave of a half wavelength on the coax cable. Further cancellations are observed at $L = n \cdot \frac{\lambda}{2}$, where n is an integer.

If the far end of the long coax is terminated with infinite resistance (free end), similar oscillations are seen as the frequency is varied, but in this case the lowest frequency cancellation at the oscilloscope corresponds to a standing wave of a quarter wavelength, and further cancellations are observed at $L = \left(n + \frac{1}{2}\right) \cdot \frac{\lambda}{2}$.

Knowing the f frequency of the cancellations and the above relations between cable length and wavelength at the cancellation points the speed of the signal propagation can be determined from the $v_{\text{signal}} = \lambda \cdot f$ relation.



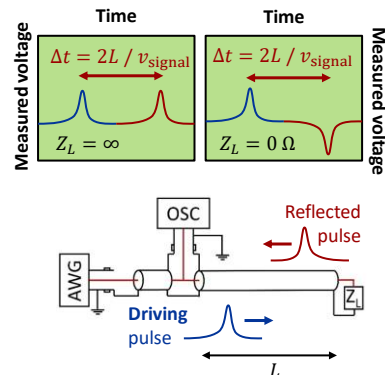
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Experiments with transmission lines: measurement of propagation speed by measuring reflected pulses

Experiment: testing cable-end reflection using pulses. Instead of sinusoidal driving we use pulses. The output pulse is seen directly at the input of the oscilloscope, but the pulse reflected from the end of the cable is also displayed with a time-delay. In the case of a 50Ω termination there is no reflected signal. In case of free wire-end ($Z_L = \infty$) the entire pulse is reflected with identical sign to the original pulse. In case of short circuit at the wire end ($Z_L = 0\Omega$) the entire pulse is reflected with opposite sign. From the time difference between the original and the reflected pulse, the propagation speed of the signal ($\sim 2/3$ speed of light) can be determined.

Note, that the $\Gamma = \frac{U^-}{U^+} = \frac{Z_L - Z_0}{Z_L + Z_0}$ formula was derived for harmonic waves. A pulse can be decomposed to harmonic waves with various circular frequencies ("Fourier transformation", see later), i.e. the above formula applies for the harmonic wave components. If the terminating impedance is a simple resistor ($Z_L = R_L$), the above formula is independent of the circular frequency, i.e. the reflected pulse does not change its shape, and the Γ reflection coefficient is valid for the pulse amplitude. For capacitive and inductive loads, however, the frequency-dependent reflection coefficients yield the strong modification of the pulse shape.

Finding contact faults in the middle of the wire: while a portion of the pulse still travels to the wire end to be reflected, partial reflections also occur at the position of the fault. The distance of the fault can be determined by the time-delay of the partial wave reflected on the fault.



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Experiments with transmission lines: the case of non-coaxial cables

Gedanken-experiment: a high-frequency signal is passed through a simple audio wall cable, or in an old flat antenna cable (see figures). The transmission of the cable is tested as a function of frequency using an oscilloscope. If the cable is placed in water, the measured voltage changes significantly. This is because the dielectric constant of the medium between the two wires changes significantly, i.e. the capacitance between the wires changes. Such a phenomenon is not observed with coaxial cables, where the capacitance between the inner core and the shielding is determined by the fixed geometry and the dielectric constant of the plastic insulation surrounding the inner core, and is not significantly affected by environmental influences.

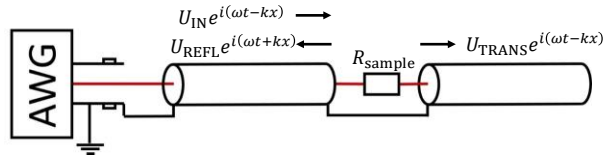
Furthermore, in a rolled old antenna cable the signal at the beginning of the cable can be directly coupled to the cable end without propagation through the entire cable length. This phenomenon also yields unwanted signal distortion.



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Transmission and reflection on an inline resistor

What happens if the resistor is not placed at the wire-end, but between the inner wires of two coaxial cables. An incoming harmonic wave with U_{IN} amplitude is sent to the resistor from the AWG. Due to the wave-nature of the signal propagation, a part of the wave is reflected on the resistor (U_{REFL}), while the other part is transmitted (U_{TRANS}).



$$U_{LEFT} = U_{IN} e^{i(\omega t - kx)} + U_{REFL} e^{i(\omega t + kx)}, \quad U_{RIGHT} = U_{TRANS} e^{i(\omega t - kx)}$$

$$I_{LEFT} = \frac{U_{IN}}{Z_0} e^{i(\omega t - kx)} - \frac{U_{REFL}}{Z_0} e^{i(\omega t + kx)}, \quad I_{RIGHT} = \frac{U_{TRANS}}{Z_0} e^{i(\omega t - kx)}$$

Current conservation at the resistor ($x=0$ position): $I_{LEFT, x=0} = I_{RIGHT, x=0} \Rightarrow U_{IN} - U_{REFL} = U_{TRANS}$

Furthermore, we apply Ohm's law for the resistor: $U_{LEFT, x=0} - U_{RIGHT, x=0} = I_{LEFT, x=0} \cdot R_{sample}$

$$\Rightarrow U_{IN} + U_{REFL} - U_{TRANS} = \frac{R_{sample} \cdot U_{TRANS}}{Z_0}$$

Merging these results (current conservation + Ohm's law): $\frac{U_{TRANS}}{U_{REFL}} = \frac{2Z_0}{R_{sample}}$

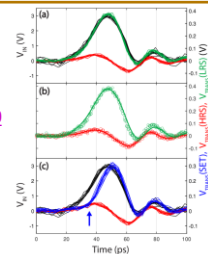
Or alternatively:

$$\frac{U_{TRANS}}{U_{IN}} = \frac{2Z_0}{R_{sample} + 2Z_0}, \quad \frac{U_{REFL}}{U_{IN}} = \frac{R_{sample}}{R_{sample} + 2Z_0}$$

The voltage drop on the sample:

$$U_{sample}(t) = 2U_{REFL} e^{i\omega t} = 2U_{IN} e^{i\omega t} \frac{R_{sample}}{R_{sample} + 2Z_0}$$

Example: world-record resistive switching in memristors:
<https://doi.org/10.1002/aelm.202201104>



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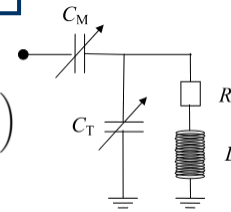
Complex impedance matching at a specific frequency

C_T sets the resonance frequency: $\omega_{res} \approx 1/\sqrt{LC_T}$

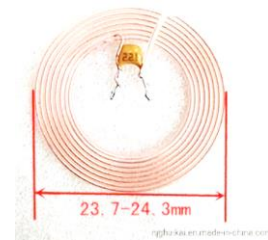
This scales the parasitic resistance up: $\text{Re } Z_L(\omega = \omega_{res}) \approx R \left(1 + \frac{L/C_T}{R^2}\right)$

By proper choice of L and C_T , the real part of the impedance can be set to 50 Ohm

The imaginary part is inductive, can be compensated by C_M



Anti-theft LC circuit, inductive coupling

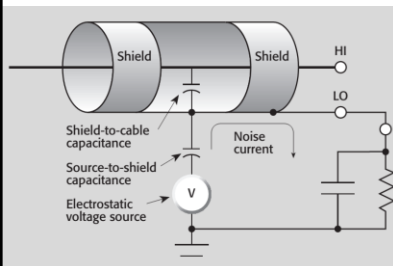
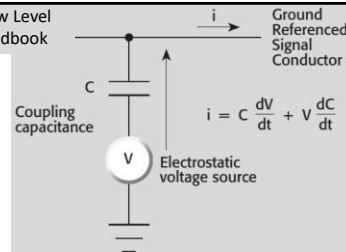


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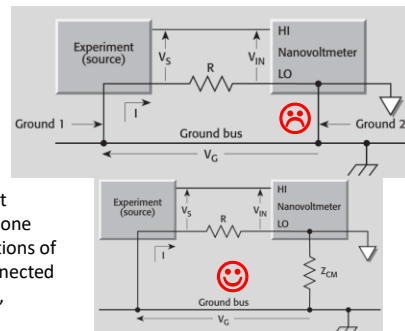
Suppressing external disturbances: electrostatic coupling, grounding

Our test leads can be capacitively coupled to external voltage sources. If the voltage or the capacitance varies with time (wires moving relative to each other), parasitic currents are generated in the circuit under test, which disturb the measurement. Causes: nearby power lines, RF fields, or even researchers walking past the measurement system :-). This causes unstable measurements, especially in high impedance circuits, where the accumulated charges are slowly discharged.

Source: Keithley Low Level Measurements Handbook



Solution: shield the circuit with a grounded metal enclosure or use shielded conductors. In this case, the currents generated by the electrostatic coupling flow through the shielding to the ground. It is also important that the cables are mechanically fixed, i.e. that the capacitance does not change over time due to mechanical vibrations.



Poor earthing can also cause problems, as there can be a voltage drop between different earth points, resulting in unwanted current flowing through the earth line. Solution: the circuit is grounded at one point (e.g. negative point of voltage source) and the other connections of the earth line (e.g. negative point of voltmeter) are floating or connected to ground through a finite resistor. The use of solid (low resistance, strongly connected) and short earth conductors is recommended.

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